WORKED SOLUTION

STEP II 2012, Question 13

Probability and Statistics • Poisson Nearest Distance

STEP II 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.19, *Random Points in the Plane*: https://beyondstep.keithmcnulty.org/ch12/spatial-poisson.html

The key idea

The event "the nearest supermarket is at least y away" is exactly the event "there are no supermarkets in the circle of radius y centred at our point", and the Poisson model tells us the probability of that directly. This gives the cumulative distribution function of Y, and differentiating gives the density. The moments are then Gaussian-type integrals: one integration by parts turns $\mathrm{E}(Y)$ into the integral we're allowed to assume, and a substitution $u = y^2$ makes $\mathrm{E}(Y^2)$ elementary.

Solution

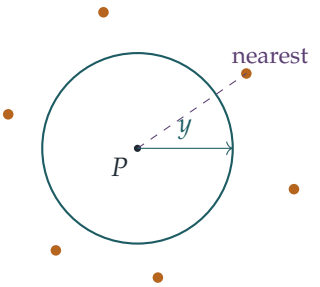
Throughout, we treat the region as so large that edge effects can be ignored, so a circle of radius y about the chosen point lies inside the region for every y that matters.

No supermarkets in a circle

Let N be the number of supermarkets inside a circle of radius y. The circle has area πy^2, so $N \sim \mathrm{Po}(\pi k y^2)$ and

$$\mathrm{P}(N = 0) = \mathrm{e}^{-\pi k y^2}.$$

The distribution of Y



Let P be the randomly chosen point. The nearest supermarket is at distance at least y from P precisely when the open circle of radius y centred at P contains no supermarkets. So, for $y \geq 0$,

$$\mathrm{P}(Y \geq y) = \mathrm{e}^{-\pi k y^2}, \qquad \text{hence} \qquad \mathrm{P}(Y < y) = 1 - \mathrm{e}^{-\pi k y^2}.$$

(Since Y is continuous, whether we write $<$ or $\leq$ makes no difference.) This is the cumulative distribution function $F(y)$ of Y, and $F(y) = 0$ for $y < 0$ because a distance can't be negative. Differentiating, for $y \geq 0$,

$$f(y) = F'(y) = \frac{\mathrm{d}}{\mathrm{d}y}\left(1 - \mathrm{e}^{-\pi k y^2}\right) = 2\pi k y\, \mathrm{e}^{-\pi k y^2},$$

which is the required density. As a quick sanity check, $F(y) \to 1$ as $y \to \infty$, so f integrates to 1.

The mean

$$\mathrm{E}(Y) = \int_0^\infty y \cdot 2\pi k y\, \mathrm{e}^{-\pi k y^2}\, \mathrm{d}y.$$

Ask yourself How can we use the given integral when there's a y^2 in front of the exponential?

Split the integrand as $y \times \left(2\pi k y\, \mathrm{e}^{-\pi k y^2}\right)$: the second factor is the derivative of $-\mathrm{e}^{-\pi k y^2}$. Integrating by parts with $u = y$,

$$\mathrm{E}(Y) = \left[-y\, \mathrm{e}^{-\pi k y^2}\right]_0^\infty + \int_0^\infty \mathrm{e}^{-\pi k y^2}\, \mathrm{d}y = \int_0^\infty \mathrm{e}^{-\pi k y^2}\, \mathrm{d}y,$$

since $y\, \mathrm{e}^{-\pi k y^2} \to 0$ as $y \to \infty$ (the exponential wins) and vanishes at $y = 0$. Now match the exponent to $-\frac{1}{2}x^2$: put $\pi k y^2 = \frac{1}{2}x^2$, that is, $y = x/\sqrt{2\pi k}$, $\mathrm{d}y = \mathrm{d}x/\sqrt{2\pi k}$. The limits stay 0 and ∞, so

$$\mathrm{E}(Y) = \frac{1}{\sqrt{2\pi k}} \int_0^\infty \mathrm{e}^{-x^2/2}\, \mathrm{d}x = \frac{1}{\sqrt{2\pi k}} \sqrt{\frac{\pi}{2}} = \frac{1}{2\sqrt{k}}.$$

The variance

For $\mathrm{E}(Y^2)$ we don't need the given integral at all, because the odd power of y lets us substitute $u = \pi k y^2$, $\mathrm{d}u = 2\pi k y\,\mathrm{d}y$:

$$\mathrm{E}(Y^2) = \int_0^\infty y^2 \cdot 2\pi k y\, \mathrm{e}^{-\pi k y^2}\,\mathrm{d}y = \int_0^\infty \frac{u}{\pi k}\,\mathrm{e}^{-u}\,\mathrm{d}u = \frac{1}{\pi k}\int_0^\infty u\,\mathrm{e}^{-u}\,\mathrm{d}u = \frac{1}{\pi k},$$

using $\int_0^\infty u\,\mathrm{e}^{-u}\,\mathrm{d}u = 1$ (integrate by parts, or recognise the mean of an exponential distribution with parameter 1). Therefore

$$\mathrm{Var}(Y) = \mathrm{E}(Y^2) - \big(\mathrm{E}(Y)\big)^2 = \frac{1}{\pi k} - \frac{1}{4k} = \frac{4-\pi}{4\pi k},$$

as required.

Checks

- *Scaling.* If supermarkets are k times as dense, lengths should shrink by a factor $\sqrt{k}$, so $\mathrm{E}(Y) \propto k^{-1/2}$ and $\mathrm{Var}(Y) \propto k^{-1}$. Both answers have this form.
- *Positivity.* $4 - \pi > 0$, so the variance is positive, as it must be.
- *Numerics.* With $k = 1$, midpoint quadrature of the density gives $\mathrm{E}(Y) = 0.5000$ and $\mathrm{Var}(Y) = 0.06831$, matching $(4-\pi)/(4\pi) \approx 0.06831$. A simulation of 10^5 random Poisson scatters of supermarkets around a point gave mean 0.4996 and variance 0.0681.

Common slips

- Writing $\mathrm{P}(Y < y) = \mathrm{e}^{-\pi k y^2}$: that is the probability of *no* supermarket within y, i.e. $\mathrm{P}(Y \geq y)$.
- Substituting $x = \sqrt{\pi k}\,y$ instead of $x = \sqrt{2\pi k}\,y$, which gives e^{-x^2} rather than the assumed $\mathrm{e}^{-x^2/2}$ and an answer out by a factor of $\sqrt{2}$.

Beyond STEP

The supermarkets here form a *Poisson point process* in the plane, the basic model of "complete spatial randomness" used for trees in a forest, stars in the sky and mobile-phone masts. The distribution of Y is a *Rayleigh distribution*, which also appears as the length of a vector whose two components are independent normal variables; the neat tail formula $\mathrm{E}(Y) = \int_0^\infty \mathrm{P}(Y \geq y)\,\mathrm{d}y$ that our integration by parts uncovered holds for every non-negative random variable. Grimmett and Welsh, *Probability: An Introduction*, covers Poisson processes and these tail-integral ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.